

WEEK 8 – STUDENT PERFORMANCE ANALYTICS DASHBOARD

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INTRODUCTION

After cleaning, transforming, and preparing the Student Performance dataset in Week 7, the prepared data was used in Week 8 to create an interactive Student Performance Analytics Dashboard using Microsoft Power BI. The dashboard was designed to analyse overall academic performance, subject-wise scores, parental education, test preparation, lunch type, grade distribution, and Pass/Fail results.

KPI cards, charts, slicers, and a drill-through Academic Profile page were created to make the dashboard interactive and easy to analyse. The Subject and Score fields from the Students Mark table were used for subject-level analysis.

Objective

To create an interactive Student Performance Analytics Dashboard in Power BI using KPI cards, charts, slicers, and drill-through functionality to analyse student academic performance and identify meaningful patterns in the data.

STEP 1 – CREATE MAIN DASHBOARD PAGE

Open the Power BI file prepared in Week 7.

Steps:

1. Open the Week 7 Power BI file.
2. Create a new report page.
3. Rename the page as **Student Performance Dashboard**.
4. Arrange the page for KPI cards, charts, and slicers.
5. Create a suitable dashboard title.

The main dashboard page provides an overview of student academic performance.

STEP 2 – CREATE KPI CARDS

Create five KPI cards to display the main academic performance indicators.

The KPI cards used are:

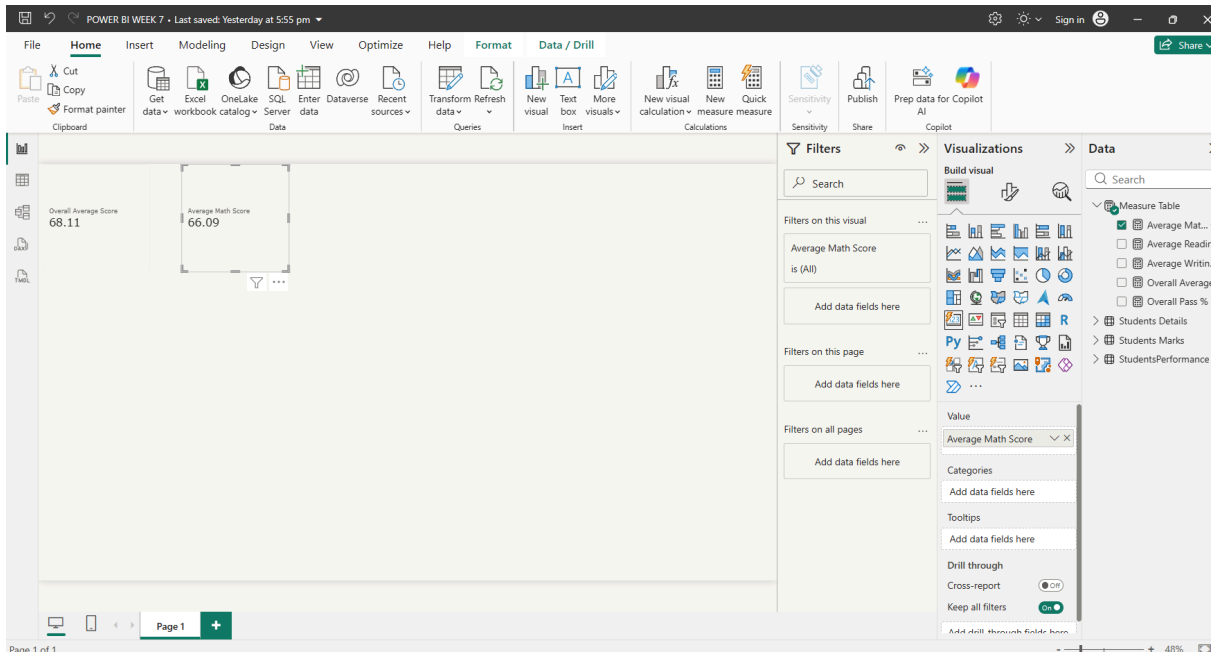
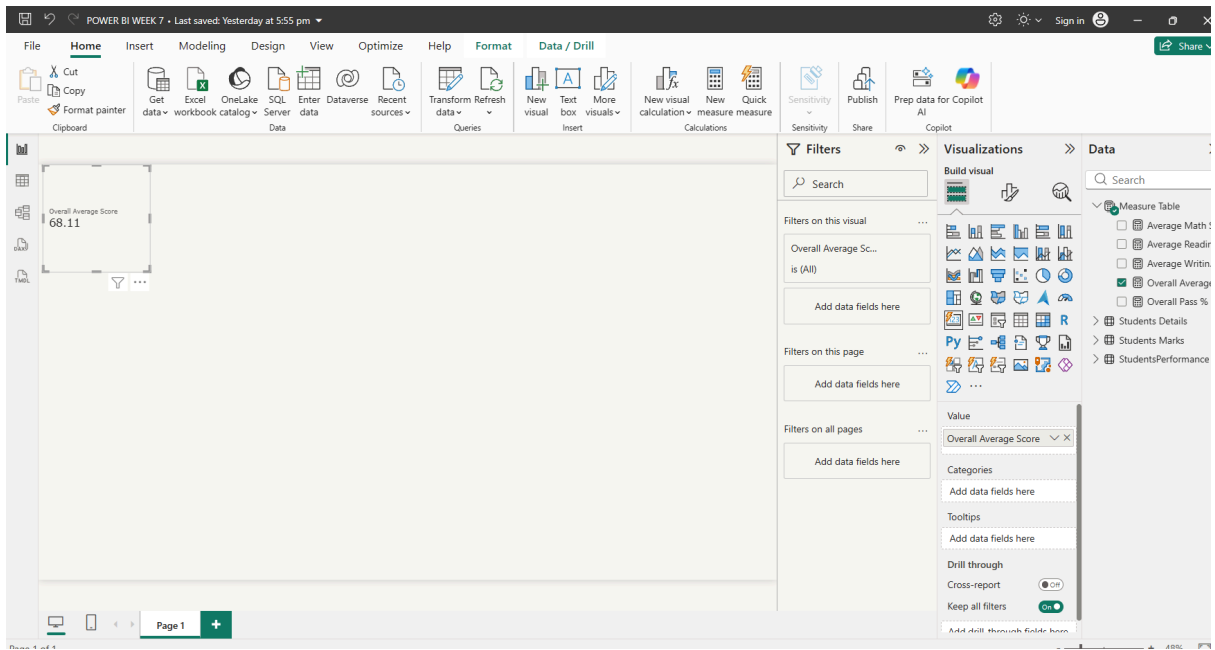
- Overall Average Score
- Average Math Score
- Average Reading Score
- Average Writing Score
- Overall Pass %

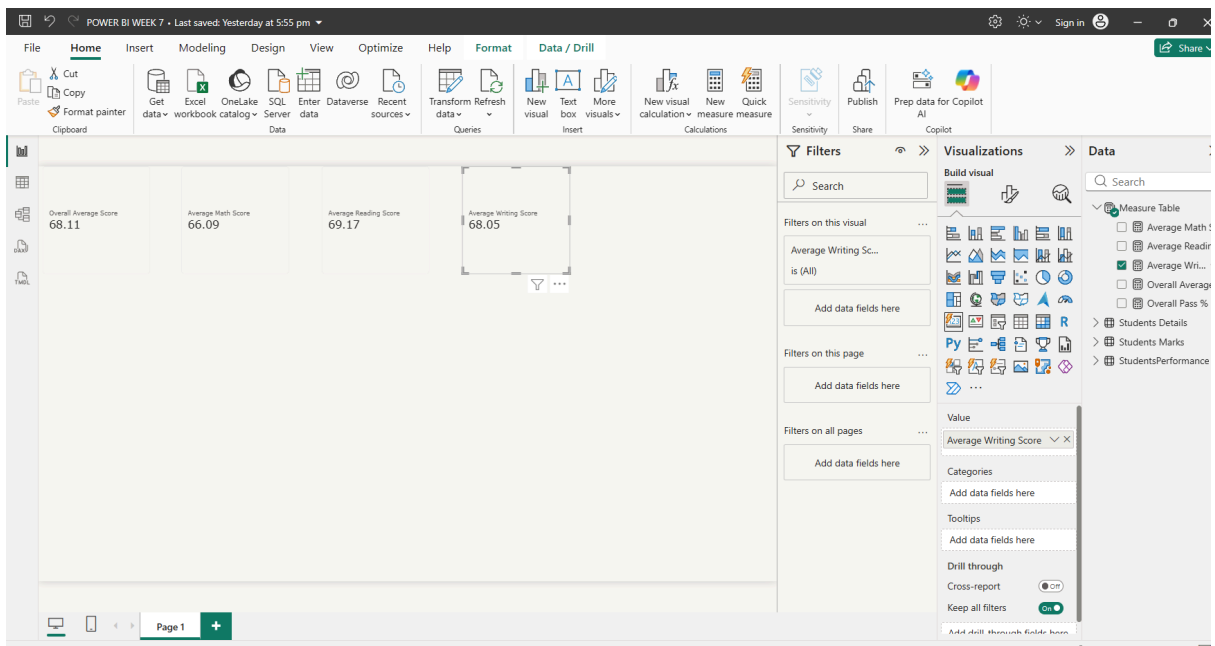
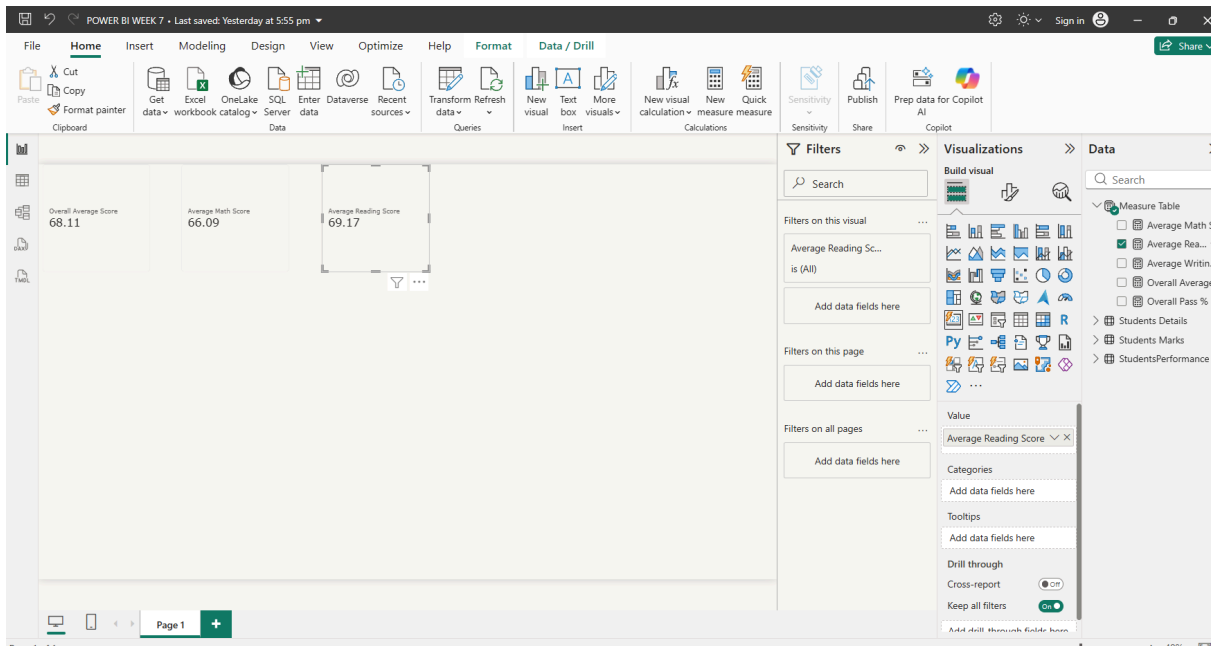
Steps:

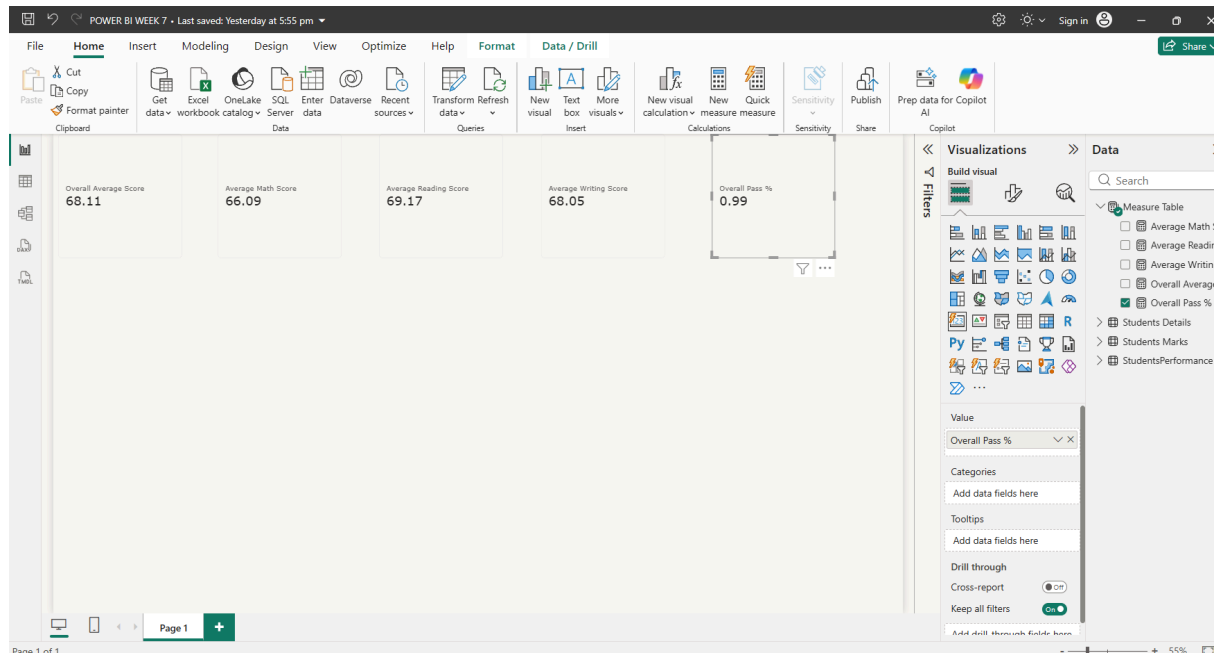
- Select the **Card** visual.
- Add the required DAX measure.
- Create separate cards for each measure.
- Arrange the five cards at the top of the dashboard.

The KPI cards provide a quick summary of the overall academic performance.

Screenshot: Five KPI cards.







STEP 3 – CREATE SUBJECT PERFORMANCE CHART

Create a chart to compare the average performance of different subjects.

Steps:

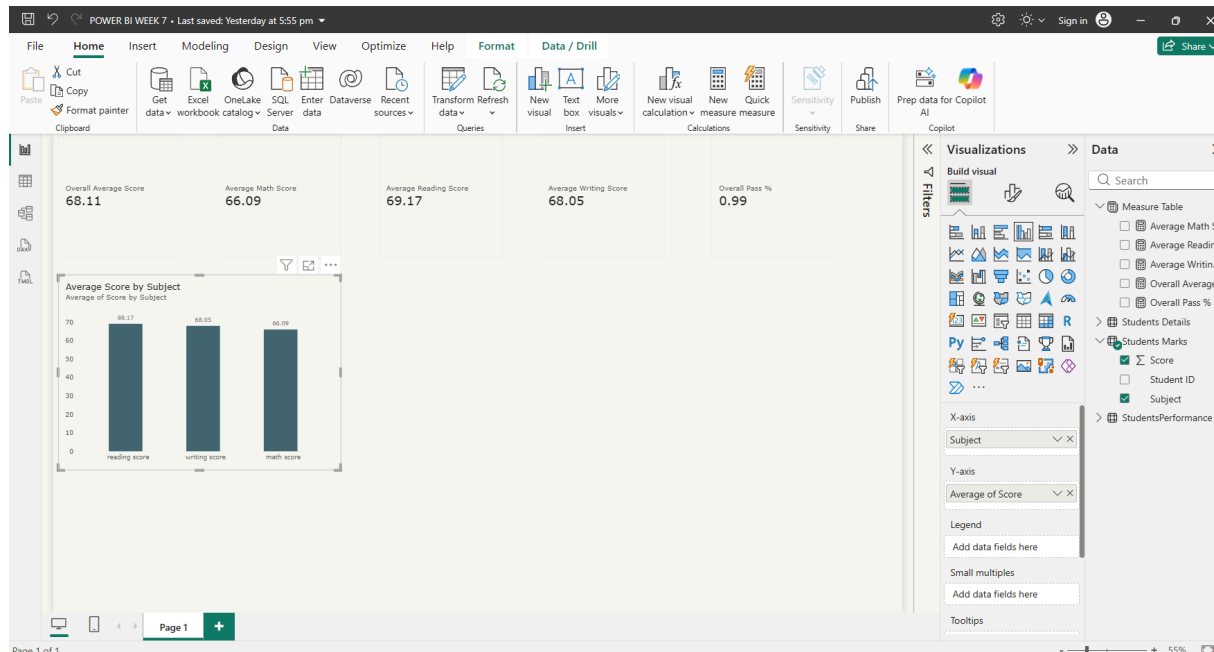
- Select a **Clustered Column Chart**.
- Add **Subject** to the X-axis.
- Add **Score** to the Y-axis.
- Change the aggregation of Score to **Average**.
- Enable Data Labels.
- Set the title as **Average Score by Subject**.

The Subject and Score fields are obtained from the Students Mark table created during Week 7.

The chart compares the average scores of:

- Math
- Reading
- Writing

Screenshot: Average Score by Subject chart.



STEP 4 – CREATE PARENTAL EDUCATION CHART

Create a chart to analyse academic performance based on parental education.

Steps:

Select a **Clustered Bar Chart**.

Add **Parental Level of Education** to the Axis.

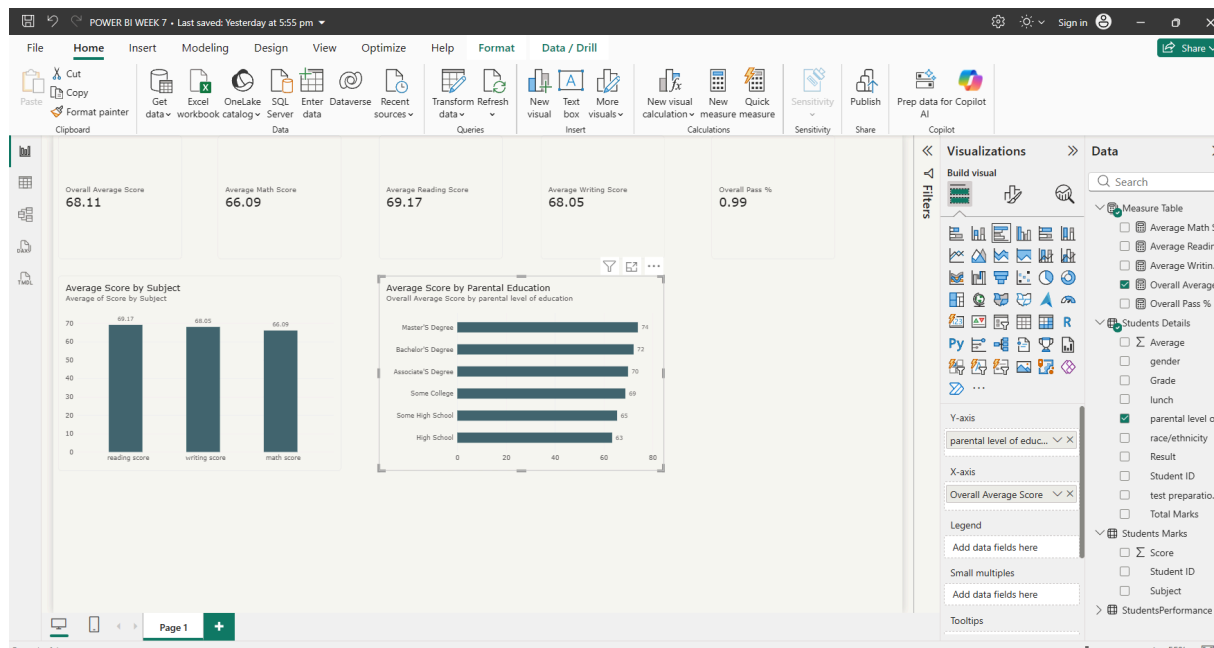
Add **Overall Average Score** to the Values.

Set the title as **Average Score by Parental Education**.

Format the chart for better readability.

This visual helps compare the average academic performance of students across different parental education levels.

Screenshot: Average Score by Parental Education chart.



STEP 5 – CREATE TEST PREPARATION CHART

Create a chart to compare students based on their test preparation status.

Steps:

Select a **Clustered Column Chart**.

Add **Test Preparation Course** to the X-axis.

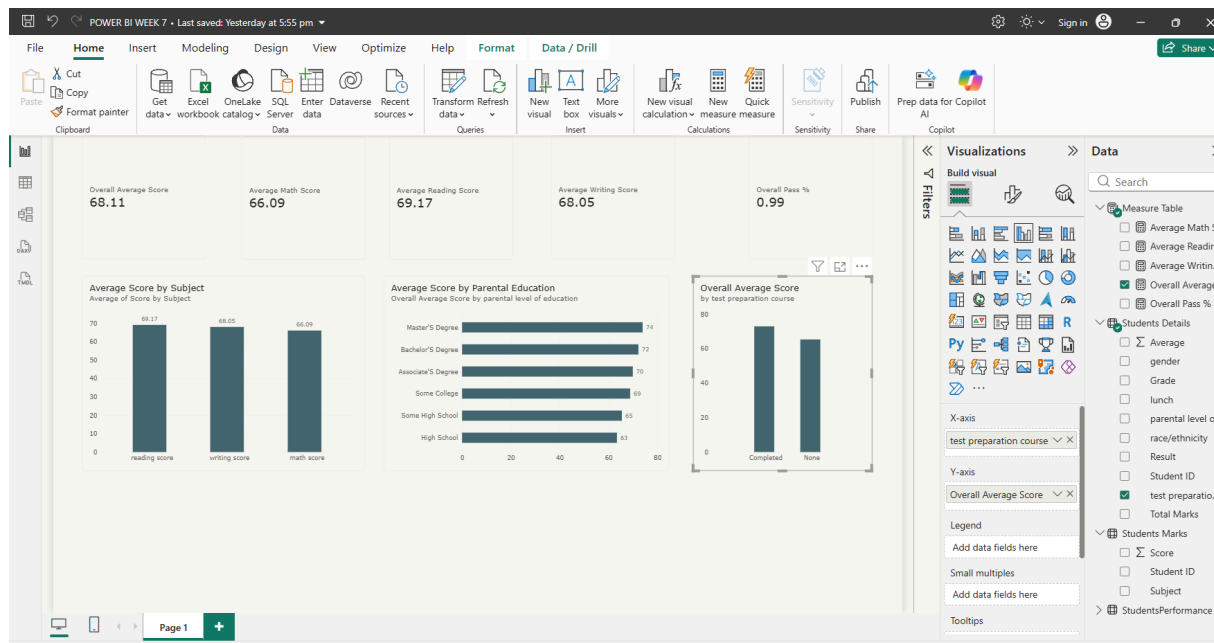
Add **Overall Average Score** to the Y-axis.

Enable Data Labels.

Set the title as **Average Score by Test Preparation Course**.

The chart compares the academic performance of students who completed the test preparation course with those who did not.

Screenshot: Average Score by Test Preparation Course chart.



STEP 6 – CREATE LUNCH TYPE CHART

Create a chart to compare academic performance based on lunch type.

Steps:

Select a **Clustered Column Chart**.

Add **Lunch** to the X-axis.

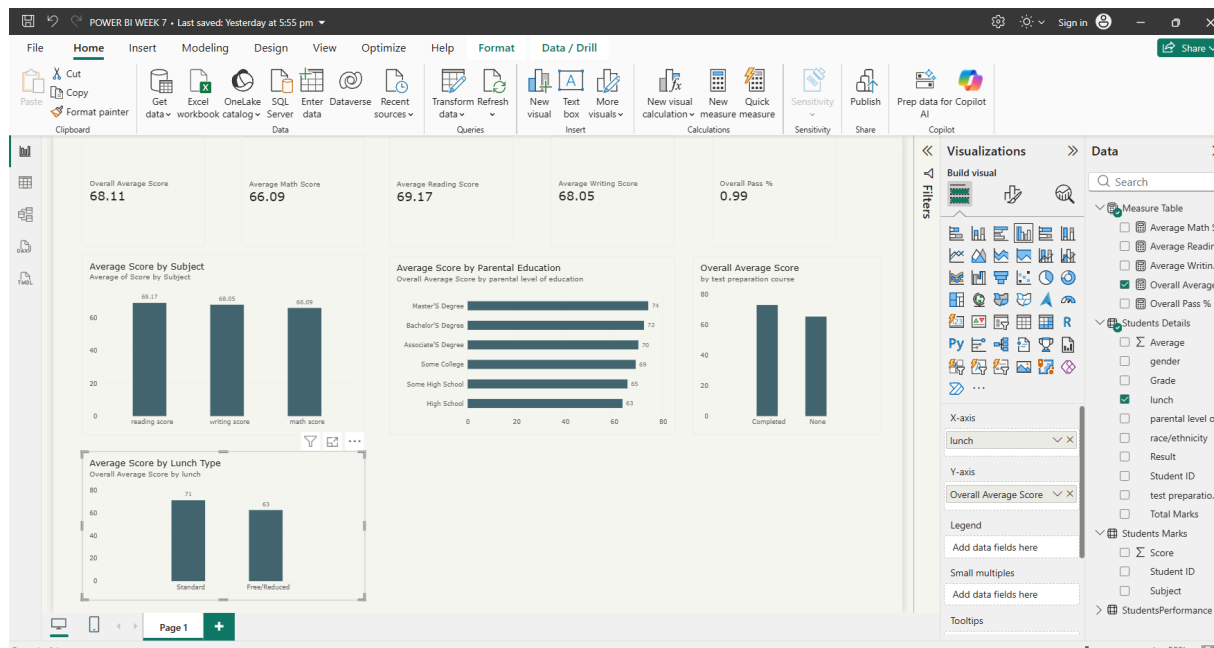
Add **Overall Average Score** to the Y-axis.

Enable Data Labels.

Set the title as **Average Score by Lunch Type**.

The chart compares the average performance of students belonging to different lunch categories.

Screenshot: Average Score by Lunch Type chart.



STEP 7 – CREATE GRADE DISTRIBUTION

Create a visual to display the distribution of students according to their grades.

Steps:

Select a **Donut Chart**.

Add **Grade** to the Legend.

Add **Student id** to Values.

Set the aggregation to Count.

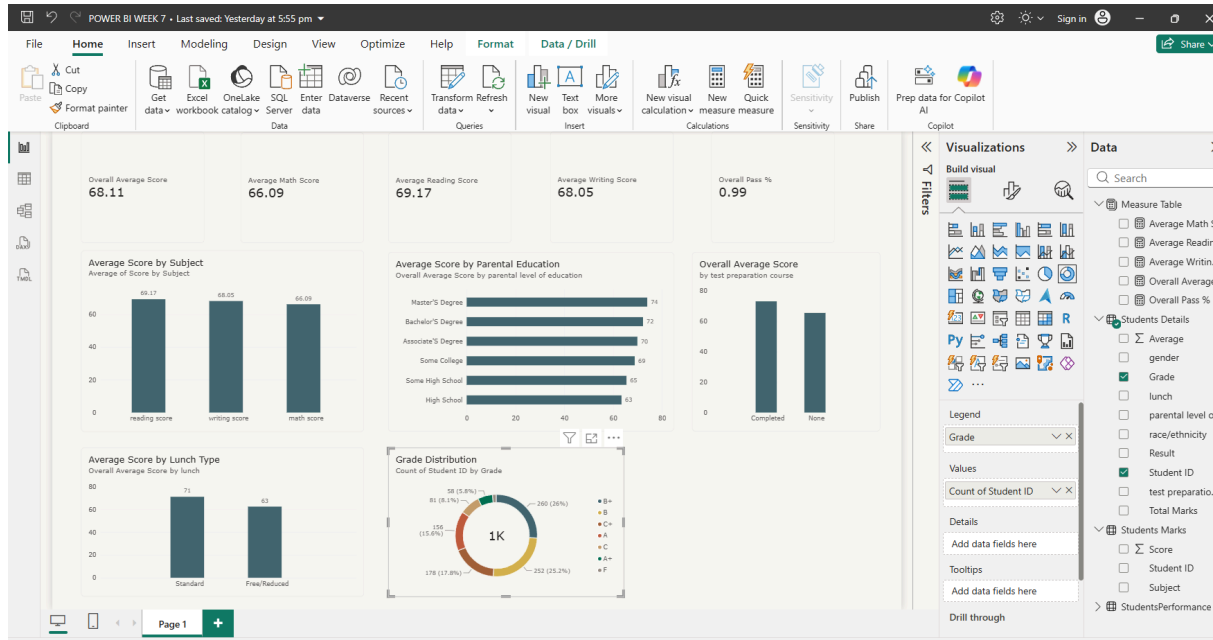
Set the title as **Grade Distribution**.

The actual grades used in the dataset are:

- A+
- A
- B+
- B
- C+
- C
- F

The chart shows how students are distributed across the different grade categories.

Screenshot: Grade Distribution chart.



STEP 8 – CREATE PASS/FAIL DISTRIBUTION

Create a visual to display the student results.

Steps:

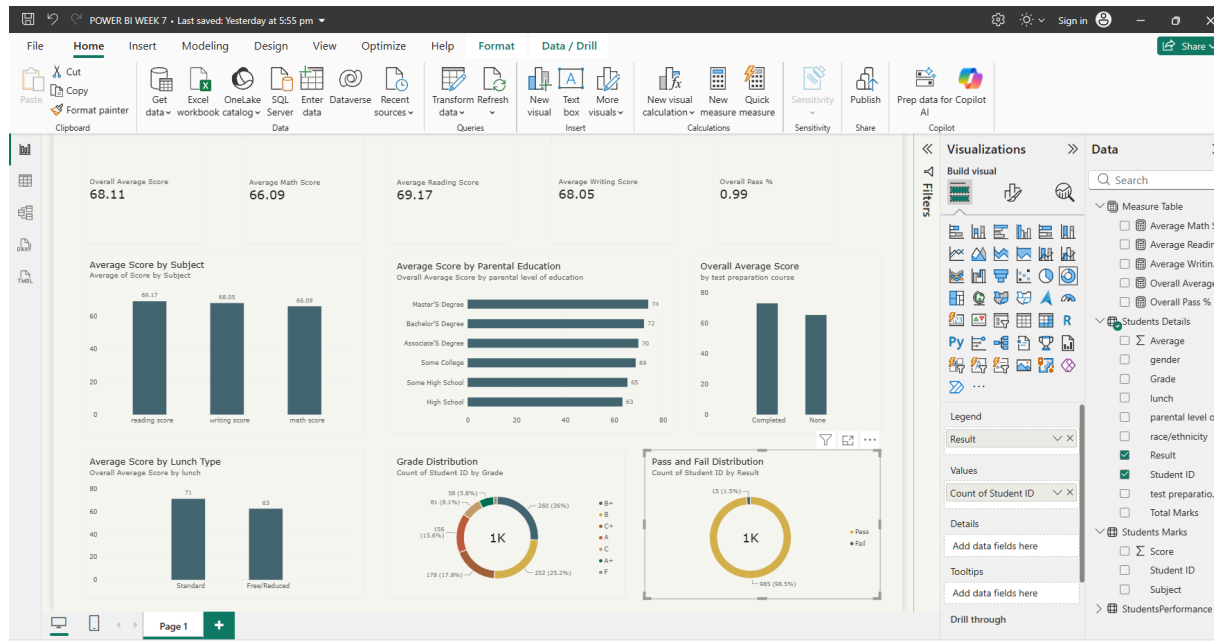
- Select a **Donut Chart**.
- Add **Result** to the Legend.
- Add **Student id** to Values.
- Set the aggregation to Count.
- Set the title as **Pass and Fail Distribution**.

The Result field contains:

- Pass
- Fail

This visual provides an overview of student Pass/Fail outcomes.

Screenshot: Pass and Fail Distribution chart.



STEP 9 – CREATE GENDER SLICER

Add a slicer for Gender.

Steps:

Select the **Slicer** visual.

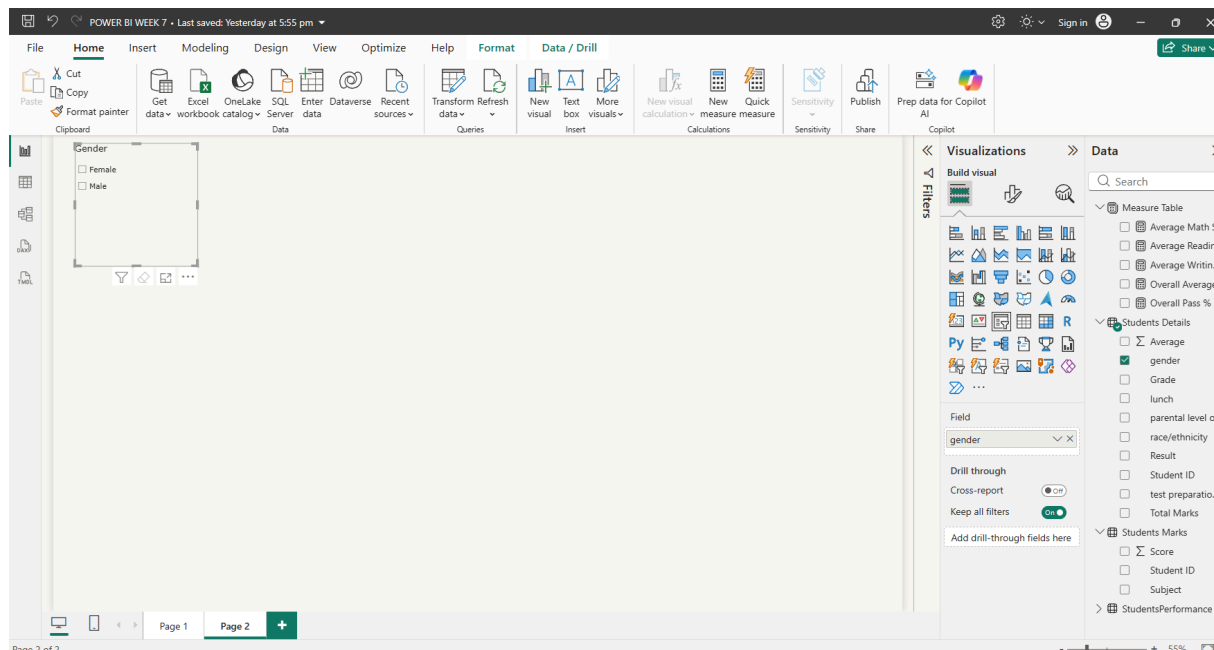
Add **Gender** to the Field.

Place the slicer on the dashboard.

Check that selecting a gender filters the dashboard visuals.

The Gender slicer allows the dashboard to be analysed for different gender categories.

Screenshot: Gender slicer.



STEP 10 – CREATE RACE/ETHNICITY SLICER

Add a slicer for Race/Ethnicity.

Steps:

Select the **Slicer** visual.

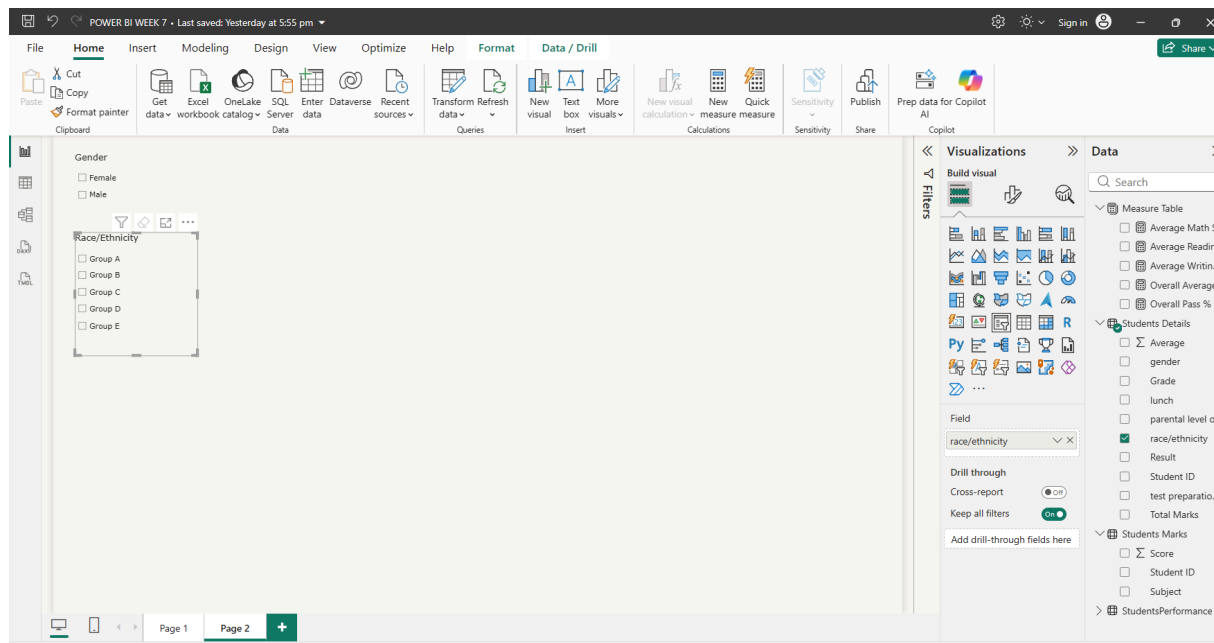
Add **Race/Ethnicity** to the Field.

Place it neatly on the dashboard.

Test the slicer by selecting different categories.

This slicer allows analysis of academic performance across different race/ethnicity groups.

Screenshot: Race/Ethnicity slicer.



STEP 11 – CREATE PARENTAL EDUCATION SLICER

Add a slicer for parental education.

Steps:

Select the **Slicer** visual.

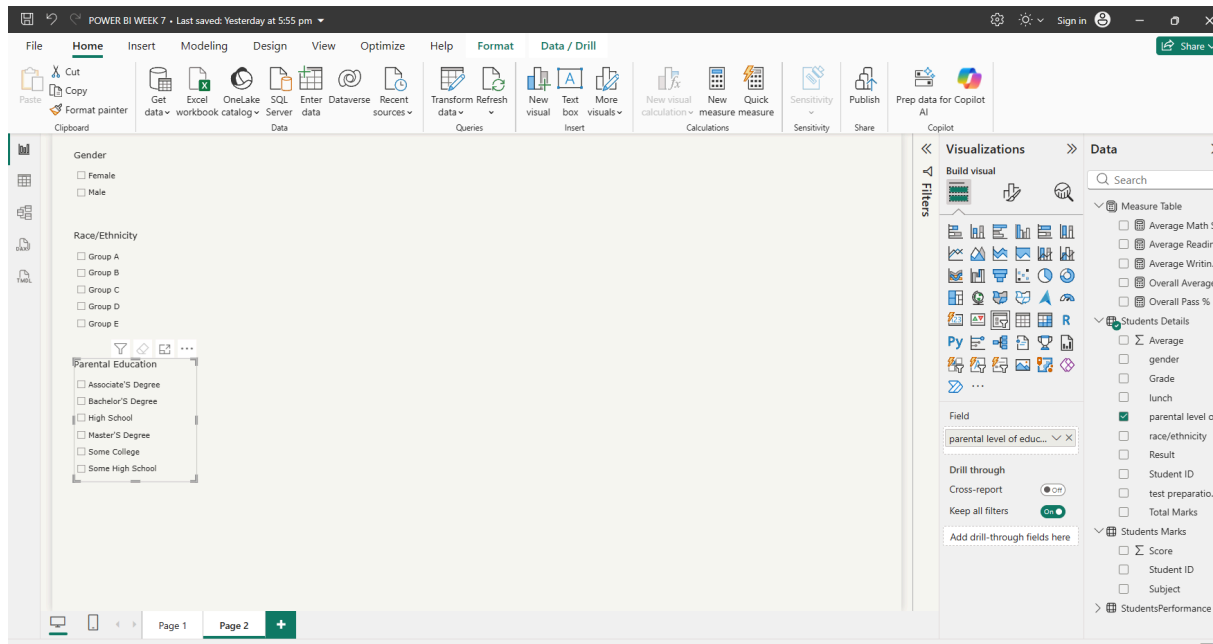
Add **Parental Level of Education** to the Field.

Place it on the dashboard.

Test the filtering interaction.

This slicer allows the user to analyse the dashboard based on parental education.

Screenshot: Parental Education slicer.



STEP 12 – CREATE LUNCH SLICER

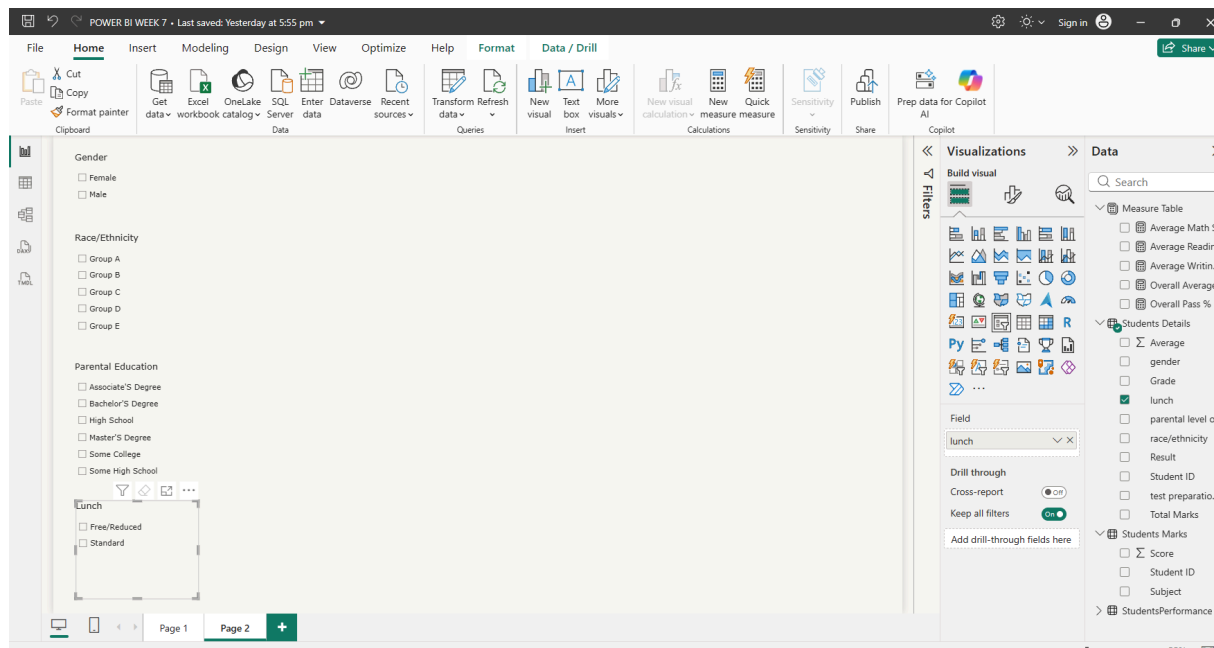
Add a slicer for Lunch.

Steps:

- Select the **Slicer** visual.
- Add **Lunch** to the Field.
- Place it on the dashboard.
- Test the filtering interaction.

This slicer allows comparison of student performance based on lunch type.

Screenshot: Lunch slicer.



STEP 13 – CREATE TEST PREPARATION SLICER

Add a slicer for Test Preparation Course.

Steps:

Select the **Slicer** visual.

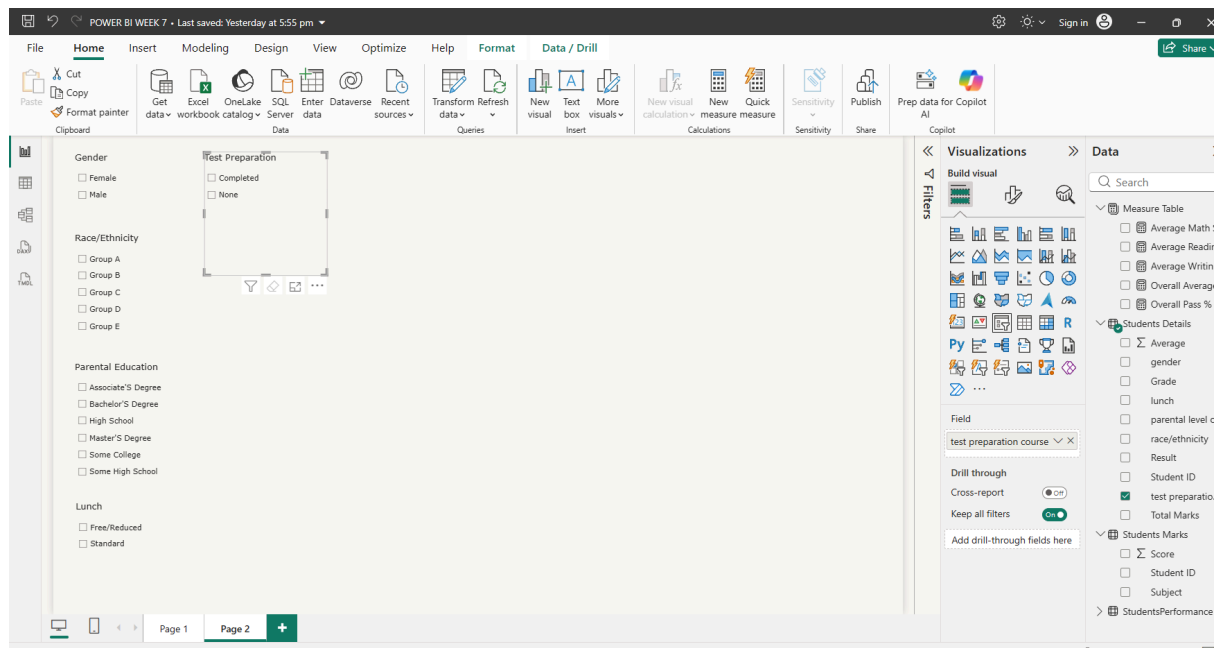
Add **Test Preparation Course** to the Field.

Place it on the dashboard.

Test the filtering interaction.

This slicer allows users to analyse students according to their test preparation status.

Screenshot: Test Preparation Course slicer.



STEP 14 – CREATE SUBJECT SLICER

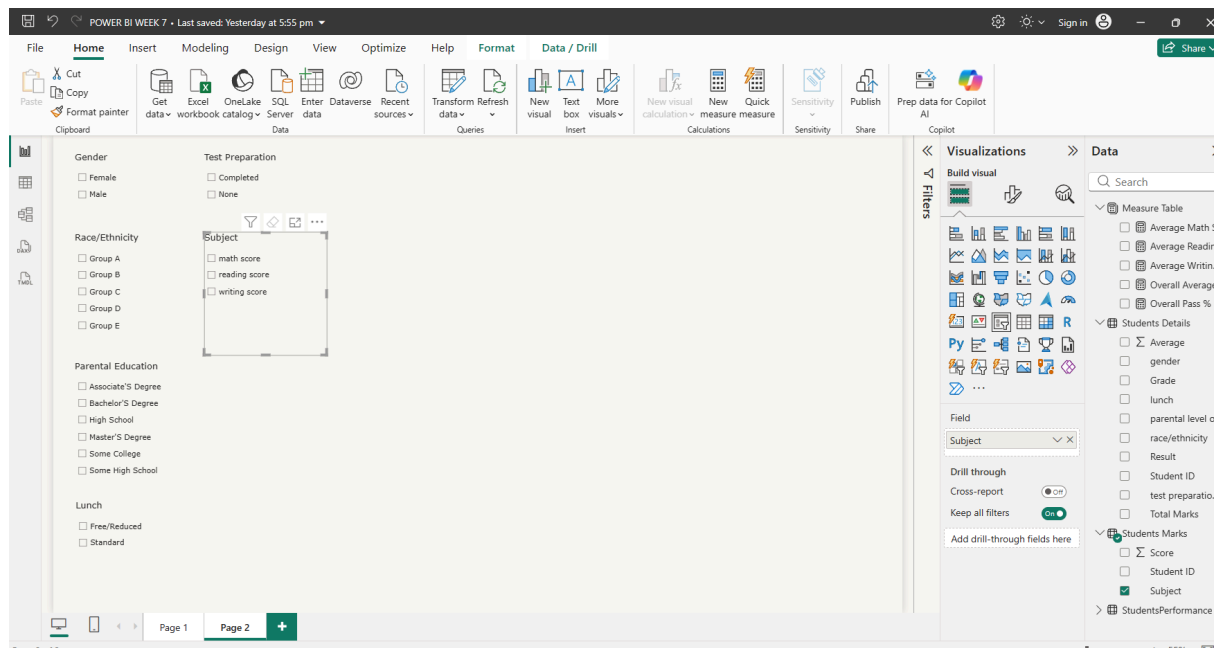
Add a slicer for Subject.

Steps:

- Select the **Slicer** visual.
- Add **Subject** to the Field.
- Place it on the dashboard.
- Test the filtering interaction.

This slicer allows users to filter the dashboard based on Math, Reading, or Writing.

Screenshot: Subject slicer.



STEP 15 – ARRANGE THE DASHBOARD

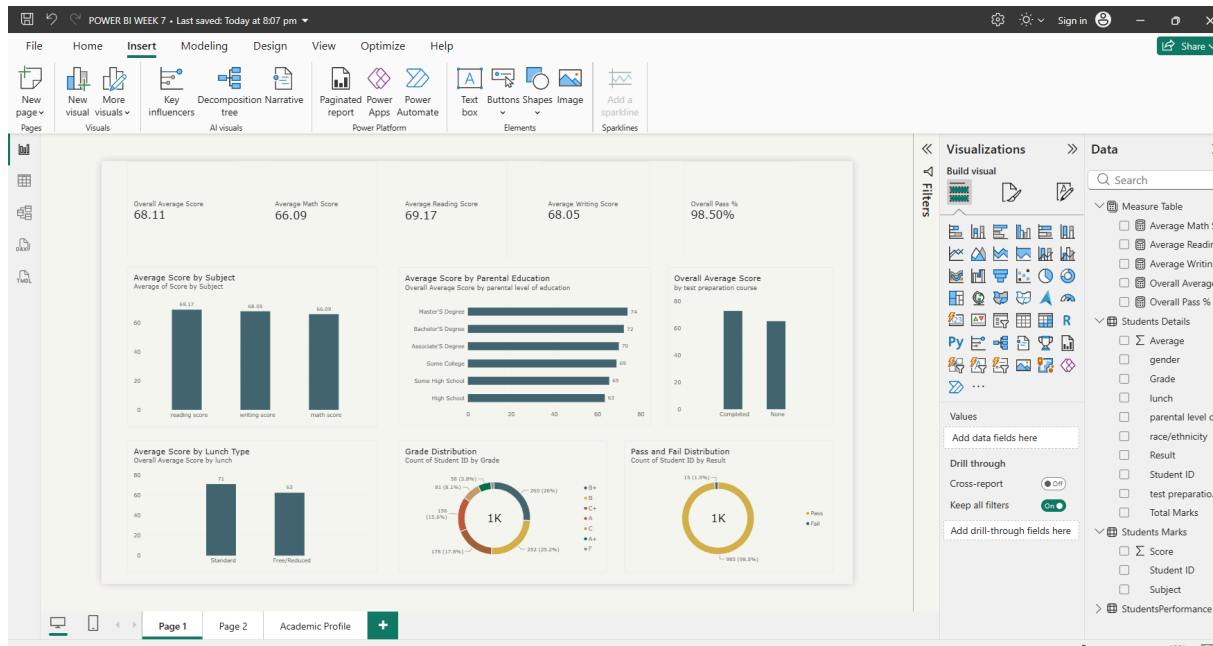
Arrange all the created visuals properly on the dashboard.

Steps:

- Place the KPI cards at the top.
- Arrange the main charts below the KPI cards.
- Place the slicers in an easily accessible area.
- Align the charts properly.
- Maintain equal spacing between visuals.
- Ensure that the dashboard is easy to read.

The dashboard layout was arranged to provide a clear visual hierarchy.

Screenshot: Arranged Student Performance Dashboard.



STEP 16 – FORMAT THE DASHBOARD

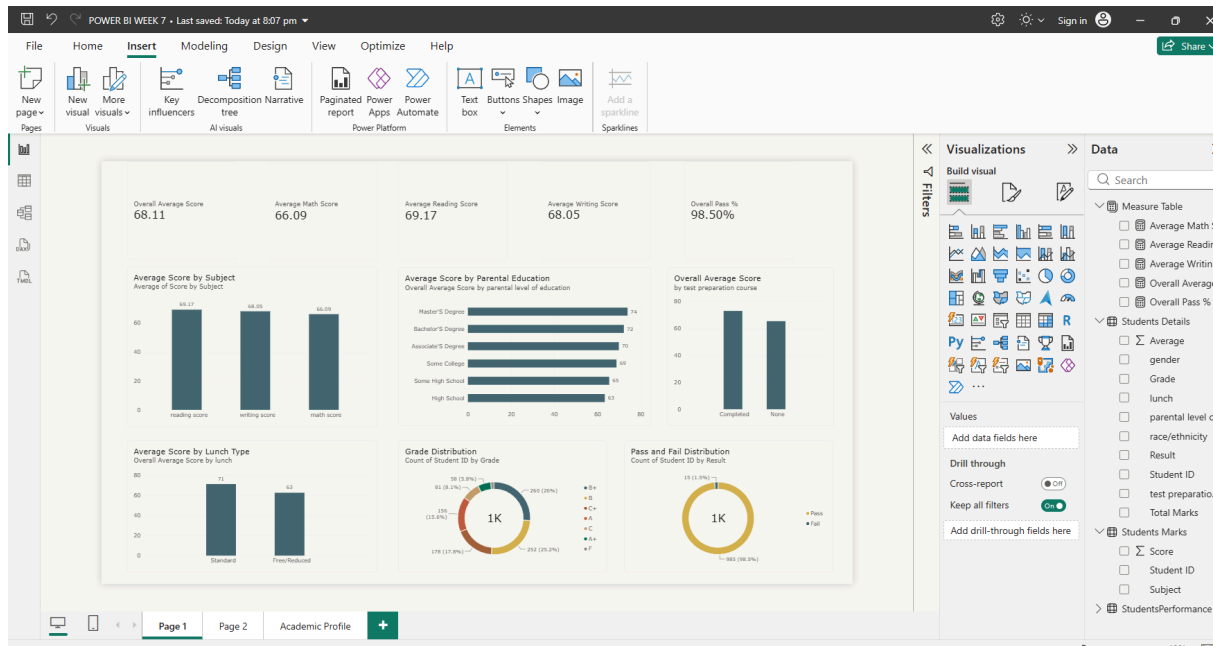
Format the dashboard to improve its appearance and readability.

Steps:

- Add meaningful titles to all visuals.
- Format the KPI cards.
- Enable data labels where required.
- Adjust font sizes.
- Align the visuals.
- Maintain consistent formatting.
- Remove unnecessary visual elements.

The formatting improves the readability and presentation of the dashboard.

Screenshot: Formatted Student Performance Dashboard.



STEP 17 – CREATE ACADEMIC PROFILE PAGE

Create a separate page for detailed student-level analysis.

Steps:

Create a new report page.

Rename the page as **Academic Profile**.

Add the required student-level visuals.

Prepare the page for drill-through functionality.

The Academic Profile page provides detailed information about an individual student.

Screenshot: Academic Profile page.

STEP 18 – ADD STUDENT ID AS DRILL-THROUGH FIELD

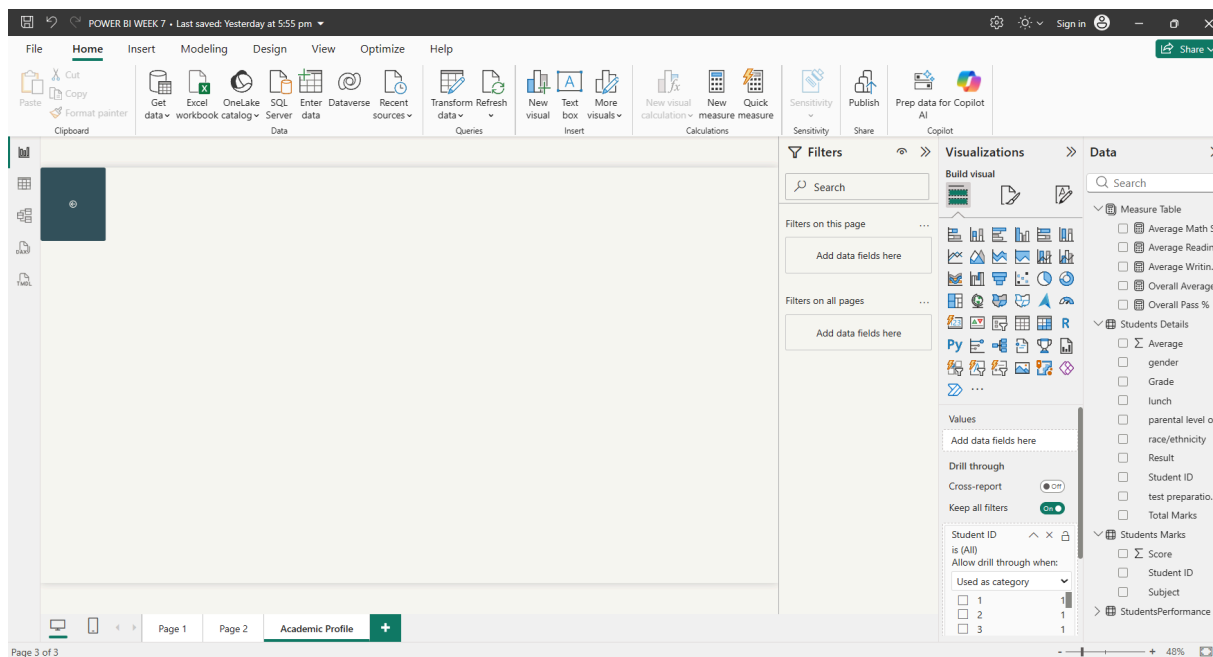
Configure the Academic Profile page for drill-through.

Steps:

Go to the **Academic Profile** page.
Locate the **Drill-through** section.
Add **Student id** as the drill-through field.
Keep **Keep all filters** enabled.

This allows the user to select a student from the main dashboard and open a detailed Academic Profile for that student.

Screenshot: Student id drill-through field.



STEP 19 – CREATE AVERAGE SCORE CARD

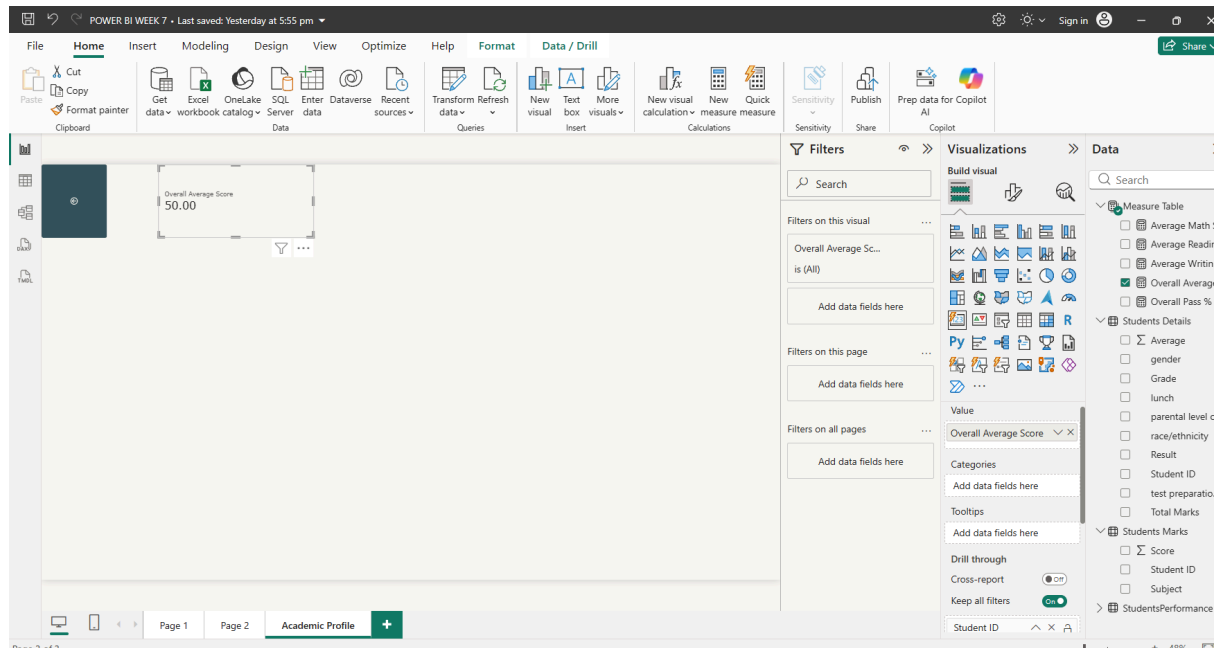
Create a card to display the selected student's average score.

Steps:

Select the **Card** visual.
Add **Overall Average Score**.
Place the card on the Academic Profile page.
Give it a suitable title.

The card displays the average score of the selected student.

Screenshot: Average Score card.



STEP 20 – CREATE GRADE CARD

Create a card to display the selected student's grade.

Steps:

Select the Card visual.

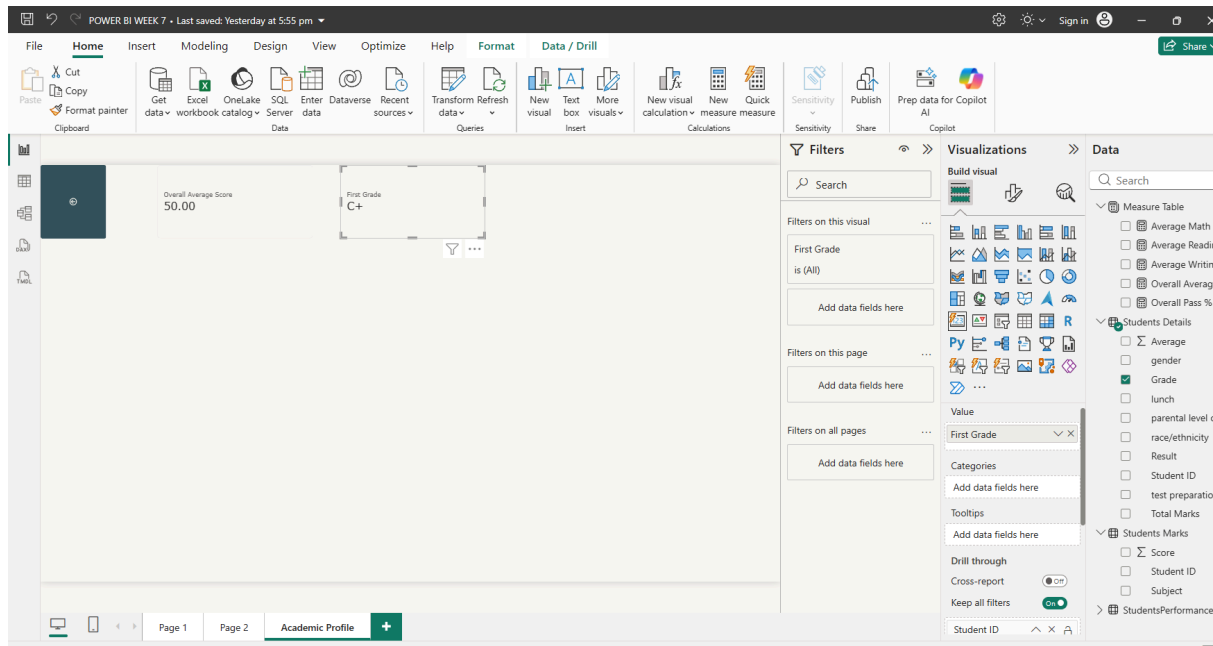
Add **Grade**.

Place it on the Academic Profile page.

Format the card appropriately.

The Grade card displays the performance grade of the selected student.

Screenshot: Grade card.



STEP 21 – CREATE RESULT CARD

Create a card to display the selected student's result.

Steps:

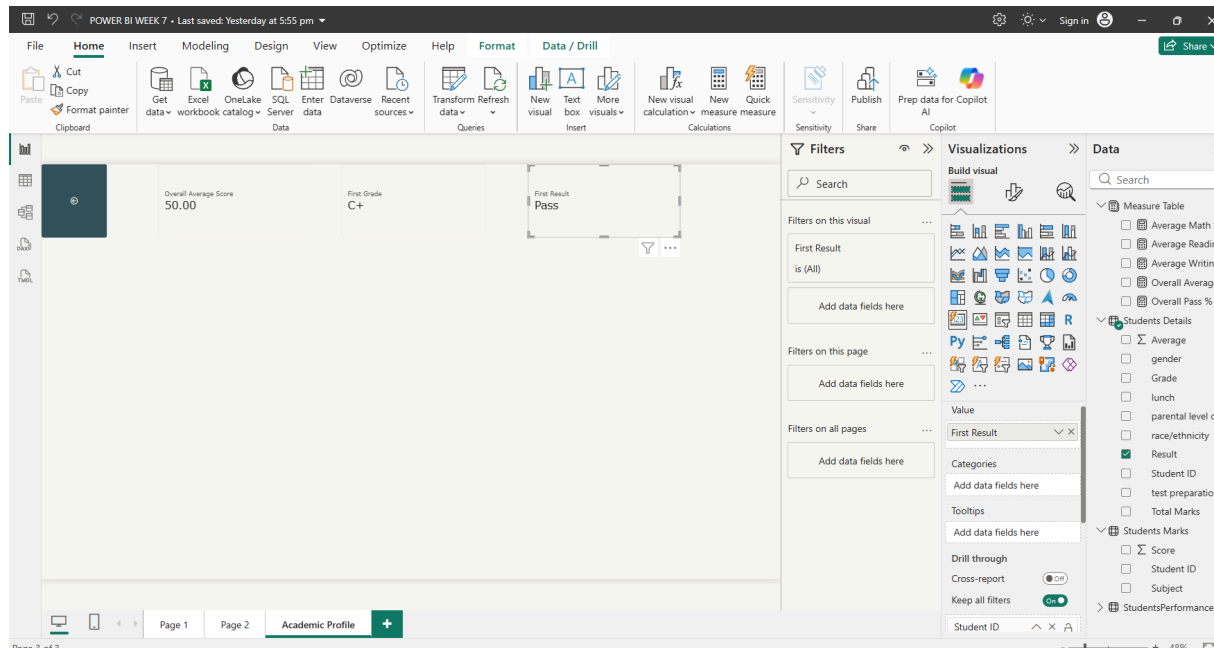
Select the Card visual.

Add **Result**.

Place it on the Academic Profile page.

The Result card displays whether the selected student has passed or failed.

Screenshot: Result card.



STEP 22 – CREATE TEST PREPARATION CARD

Create a card to display the selected student's test preparation status.

Steps:

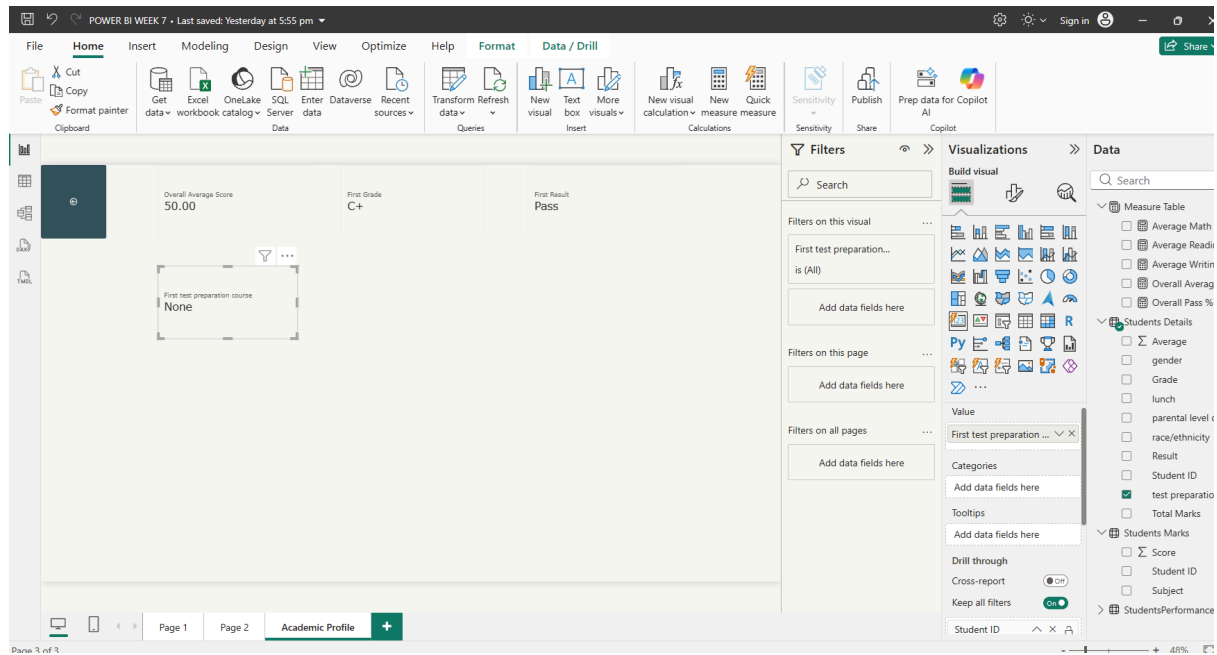
Select the Card visual.

Add **Test Preparation Course**.

Place the card on the Academic Profile page.

This card displays whether the selected student completed the test preparation course or not.

Screenshot: Test Preparation Course card.



STEP 23 – CREATE STUDENT SUBJECT PERFORMANCE CHART

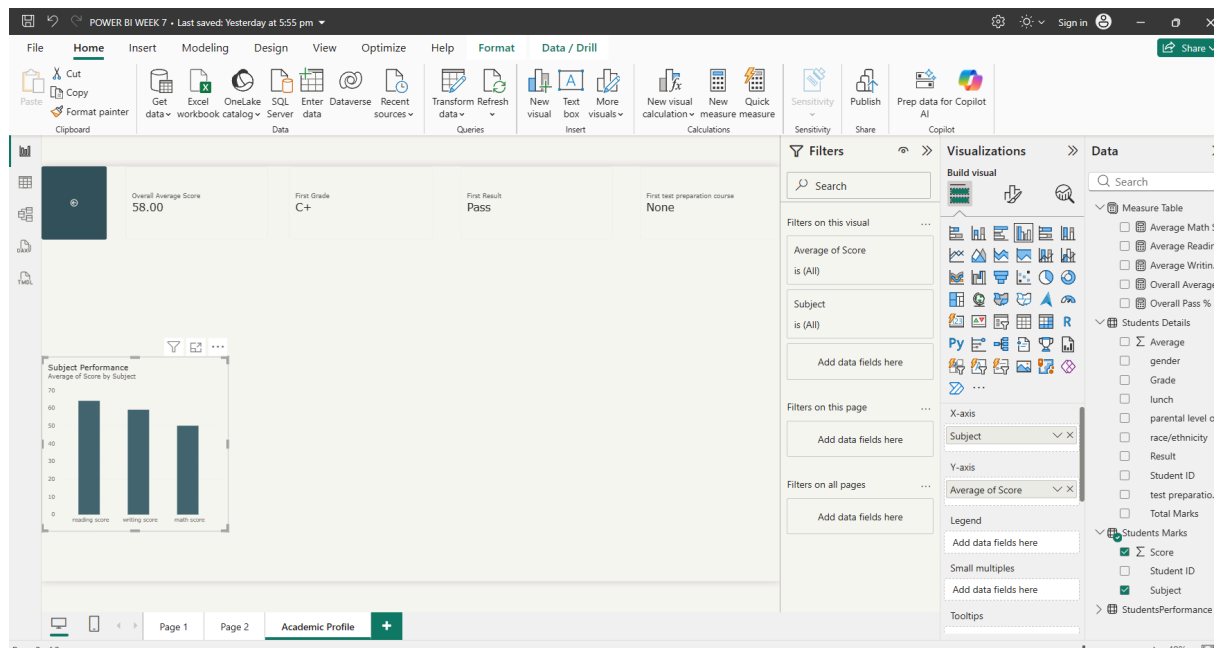
Create a chart to display the selected student's subject-wise performance.

Steps:

- Select a **Clustered Column Chart**.
- Add **Subject** to the X-axis.
- Add **Score** to the Y-axis.
- Set Score aggregation to **Average**.
- Set the title as **Student Subject Performance**.

The chart displays the selected student's performance in Math, Reading, and Writing.

Screenshot: Student Subject Performance chart.



STEP 24 – ADD BACK BUTTON

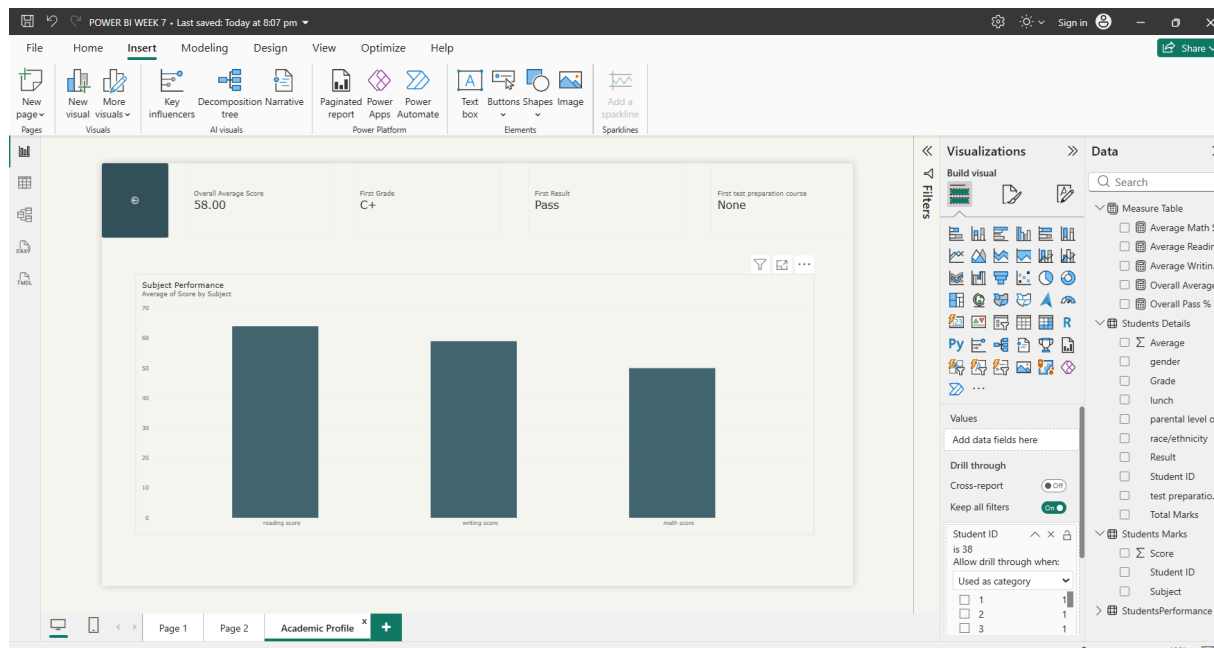
Add a Back button to the Academic Profile page.

Steps:

- Go to Insert → Buttons.
- Select the **Back** button.
- Place it on the Academic Profile page.
- Check the button action.

The Back button allows the user to return to the main dashboard after viewing the student profile.

Screenshot: Back button on Academic Profile.



STEP 25 – TEST DRILL-THROUGH FUNCTIONALITY

Test the drill-through feature.

Steps:

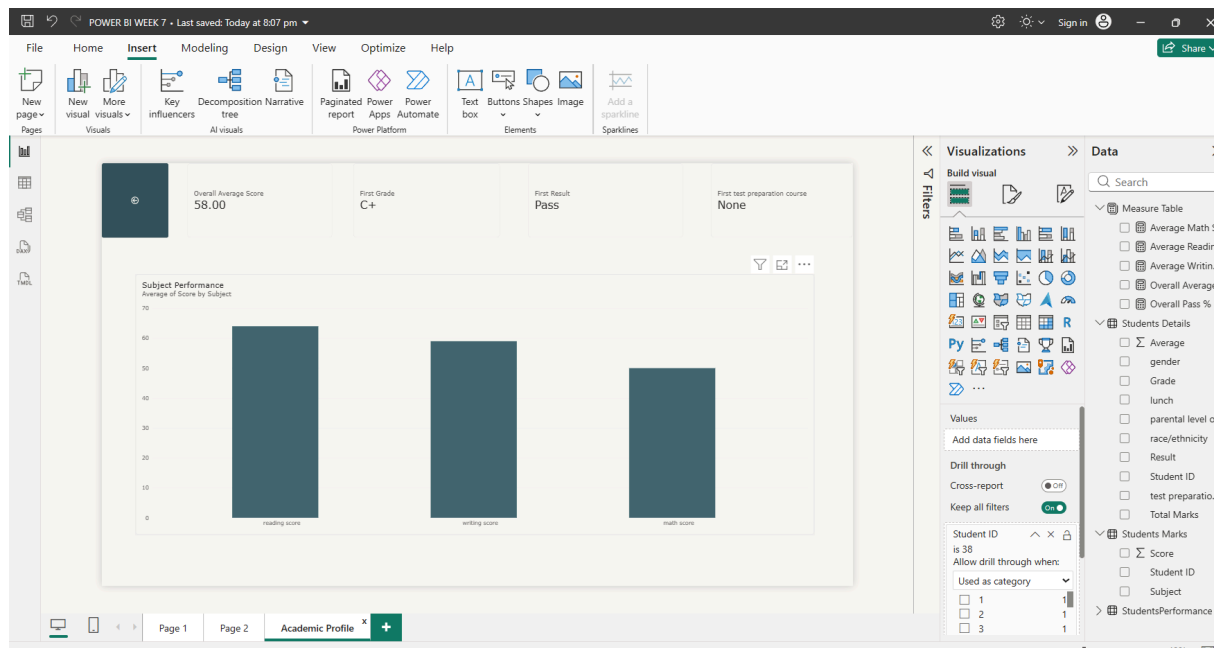
- Go to the main Student Performance Dashboard.
- Select a student using the Student ID field.
- Right-click and select **Drill through** → **Academic Profile**.
- Check the displayed student information.

Verify:

- Average Score
- Grade
- Result
- Test Preparation Course
- Subject Performance

The drill-through feature provides detailed student-level analysis.

Screenshot: Drill-through Academic Profile result.



STEP 26 – TEST SLICERS AND INTERACTIONS

Test all dashboard slicers.

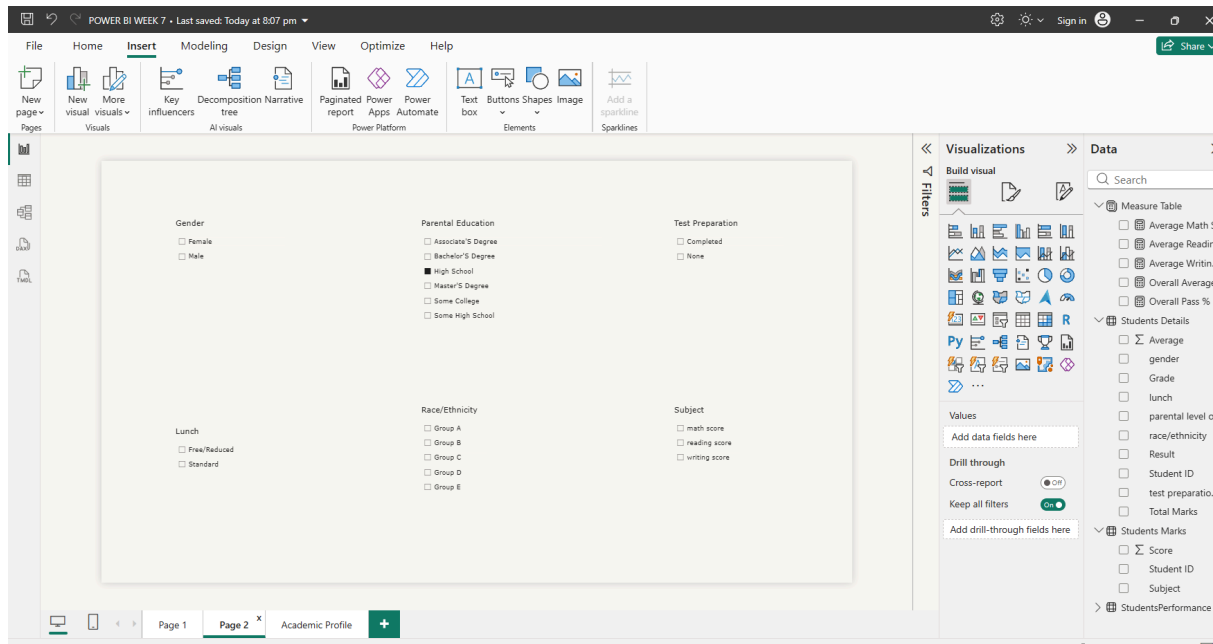
Check:

- Gender
- Race/Ethnicity
- Parental Education
- Lunch
- Test Preparation Course
- Subject

Select different values from each slicer and verify that the dashboard visuals respond correctly.

This ensures that the dashboard is interactive and functional.

Screenshot: Dashboard with slicer selection.



STEP 27 – FINAL DASHBOARD VALIDATION

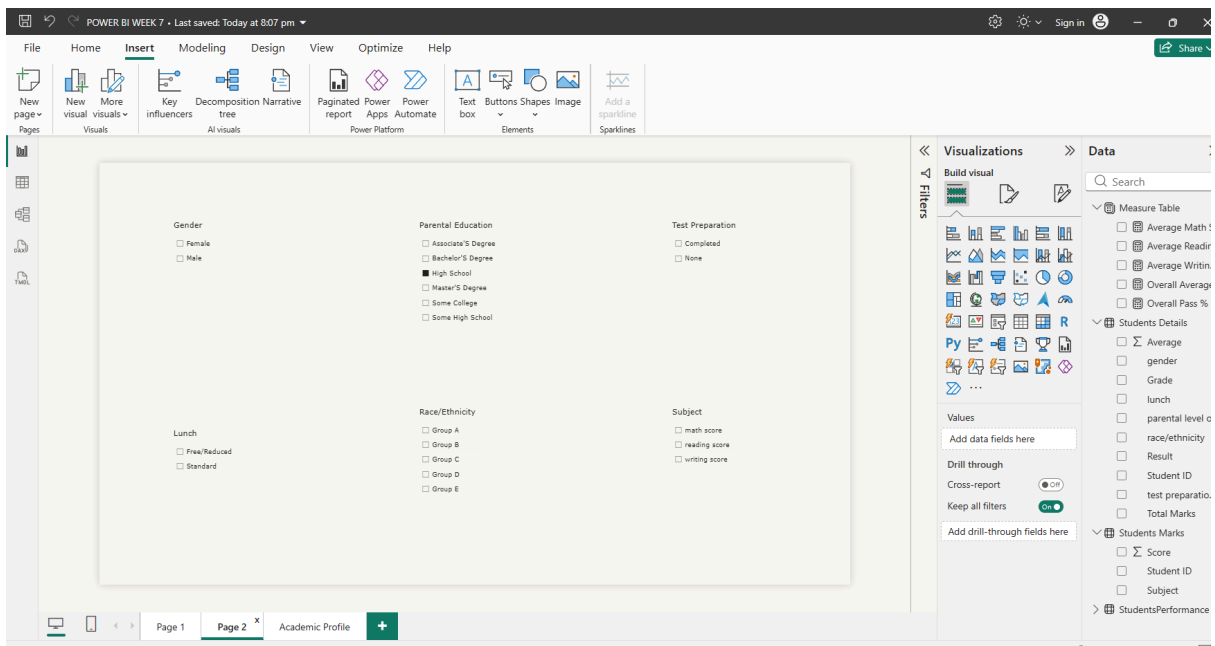
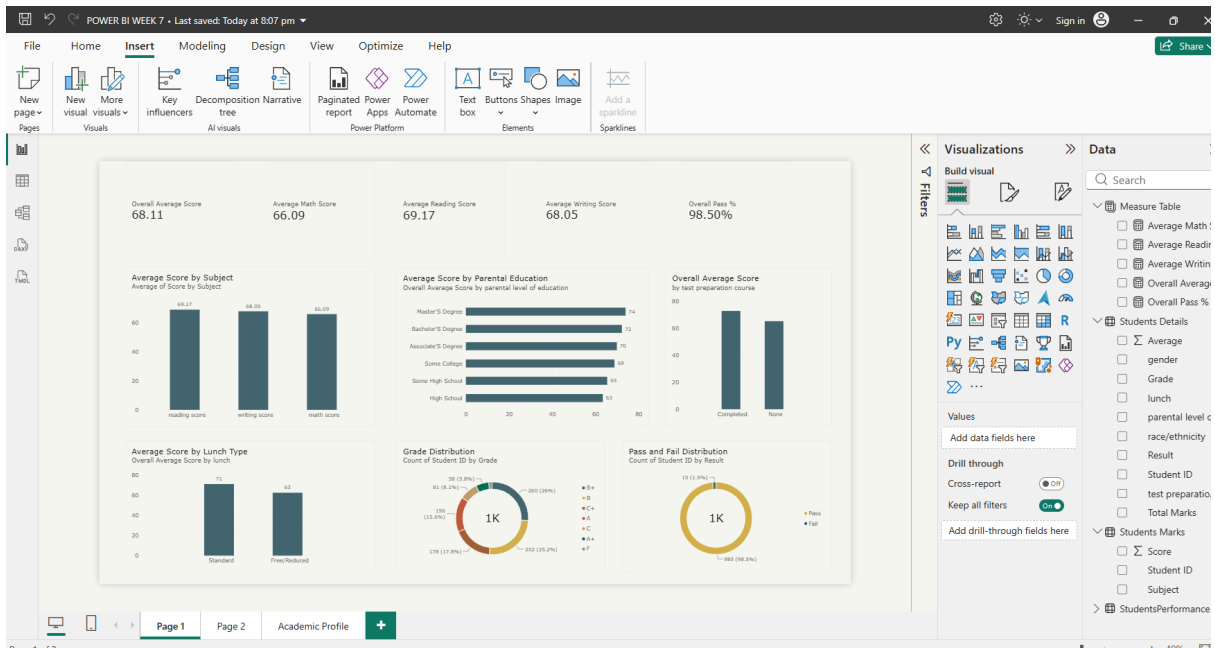
Finally, validate the completed dashboard.

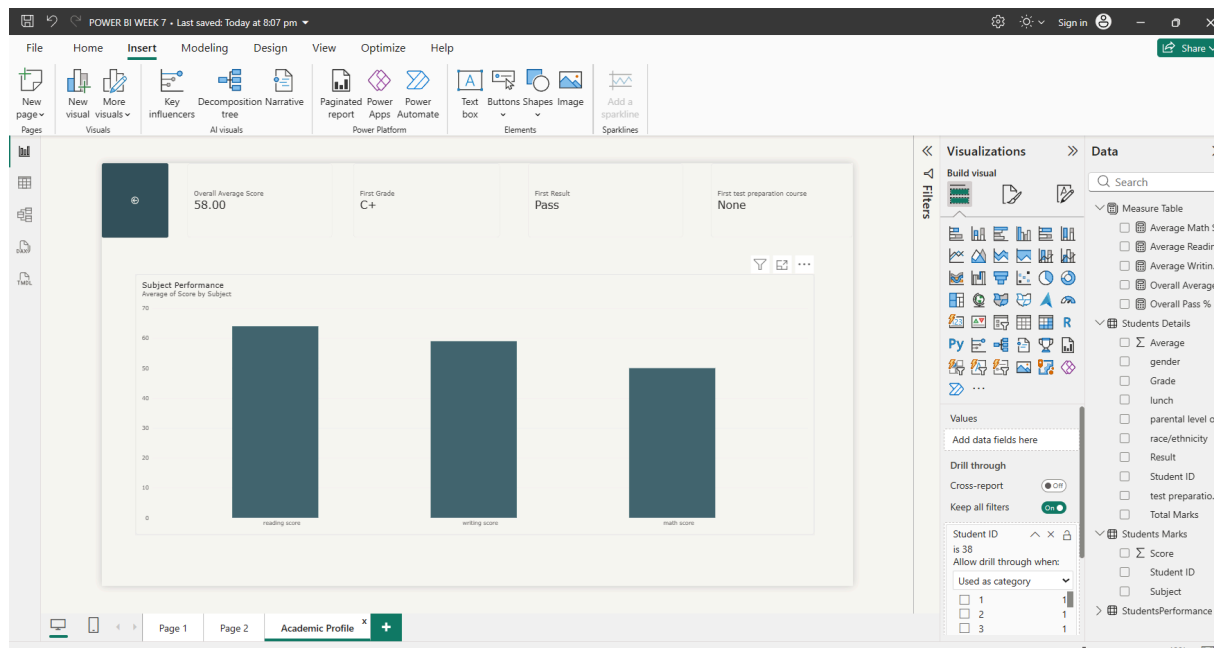
Check the following:

- Check all five KPI cards.
- Check the Subject Performance chart.
- Check the Parental Education chart.
- Check the Test Preparation chart.
- Check the Lunch Type chart.
- Check the Grade Distribution.
- Check the Pass/Fail Distribution.
- Check all six slicers.
- Check the Academic Profile drill-through.
- Check the Back button.
- Check all titles and labels.
- Check that there are no blank or error visuals.

The validation ensures that the dashboard is complete and ready for presentation.

Screenshot: Final validated Student Performance Dashboard.





CONCLUSION

The Student Performance dataset was successfully transformed into an interactive **Student Performance Analytics Dashboard** using Power BI. KPI cards were created to display important academic performance measures, and charts were used to analyse subject performance, parental education, test preparation, lunch type, grade distribution, and Pass/Fail results.

Interactive slicers were added for Gender, Race/Ethnicity, Parental Education, Lunch, Test Preparation Course, and Subject. A separate **Academic Profile** page was created using Student ID drill-through functionality to display detailed student-level information.

The fields **Attribute** and **Value** from the unpivoted table were renamed as **Subject** and **Score** and were used for subject-wise analysis. The completed dashboard provides an interactive way to explore and understand student academic performance.